

Experimentally Infused Plant and Controller Optimization Using Iterative Design of Experiments—Theoretical Framework and Airborne Wind Energy Case Study

Nihar Deodhar

Department of Mechanical Engineering,
University of North Carolina at Charlotte,
Charlotte, NC 28223
e-mail: ndeodhar@uncc.edu

Joseph Deese

Department of Mechanical Engineering,
University of North Carolina at Charlotte,
Charlotte, NC 28223
e-mail: jdeese23@uncc.edu

Christopher Vermillion¹

Mem. ASME
Department of Mechanical Engineering,
University of North Carolina at Charlotte,
Charlotte, NC 28223
e-mail: cvermill@uncc.edu

This research presents an iterative framework for optimizing the plant and controller for complex systems by fusing expensive but valuable experiments with cheap yet less accurate simulations. At each iteration, G-optimal design is used to generate experiments and simulations within a prescribed design space that is shrunk in size after each successful iteration. The shrinking of the design space is determined through statistical characterization of a response surface model, and further shrinking is achieved at successive iterations through a numerical model correction factor that is driven by the results of experiments. An initial validation of this iterative design optimization framework was performed on an airborne wind energy (AWE) system, where tethers and an aerostat are used in place of a tower to elevate the turbine to high altitudes. Using a unique lab-scale setup for the experiments, the aforementioned iterative methodology was used to optimize the center of mass location and pitch angle set point for the airborne wind energy system. The optimum configuration yielded a substantial improvement in system responses as compared to a numerically optimized configuration. The framework was recently extended to include four variables (horizontal and vertical stabilizer areas, center of mass location, and pitch angle set point). [DOI: 10.1115/1.4037014]

1 Introduction

In numerous controlled systems, the optimal controller depends on the physical system design (i.e., the *plant*) and vice versa. In response to this reality, a significant body of literature has emerged in the past two decades on the combined plant and controller design, sometimes termed the *codesign*, of such systems. For example, a nested combined plant and controller optimization has been presented for an active suspension system in Refs. [1] and [2], whereas Ref. [3] uses the same techniques for optimization of an elevator. The combined plant and controller optimization is achieved in Refs. [4] and [5] using a sequential design approach, along with a control proxy function, to achieve optimal or near-optimal designs. References [6] and [7] use a decomposition-based codesign approach, whereas Ref. [8] uses direct transcription to tackle the combined plant and controller design problem.

The aforementioned works differ in their approaches to codesign but are similar in the sense that they all rely on a *dynamic model* of the system at hand to perform their optimization, rather than integrating experiments into the design approach. A more limited set of work, most notably [9], has demonstrated the use of experiments for plant and controller optimization. The work of Ref. [9] is entirely experimentally driven, using experiments from one iteration to identify elements of a Jacobian matrix that is used to determine the direction that the design variables need to be varied at the following iteration.

¹Corresponding author.

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Airborne wind energy (AWE) systems, which are the focus of the experiments and simulations in this work, feature complex dynamics and coupling between the plant and the controller. These systems replace conventional towers with tethers and a lifting body (an aerostat, kite, or rigid wing). Replacement of traditional tower-mounted turbines with flexible tethers and a lifting body has enabled wind energy generation at altitudes up to 600 m [10]. Over the past decade, AWE technology has moved from paper to practice, with several companies and research organizations pioneering a variety of AWE systems. Google-owned Makani Power [11] and Altaeros Energies [12] have developed AWE systems with on-board wind energy generation. Ampyx Power [13], Kitegen [14], and Windlift [15] represent just a few examples of organizations that rely on cyclic motions (high-tension crosswind spool-out and low-tension spool-in) and ground-based energy generation.

A substantial body of research focuses on *control design* for AWE systems. For example, the authors in Refs. [16] and [17] have aimed to optimize energy production for kite-based systems using crosswind flight patterns. Control design for the Altaeros Buoyant Airborne Turbine (BAT) is introduced in Ref. [18]. Net power generation for the same system is maximized in Ref. [19] by determining instantaneous optimal altitude using a switched extremum seeking algorithm. A much smaller body of literature addresses the design of both the AWE system plant and the controller, with Refs. [18] and [20] being some recent examples of such work.

Most of the literature on AWE systems focuses on numerical simulations, with some validation results presented on full-scale systems. However, AWE systems fall under the broad category of complex nonlinear systems with imperfect numerical models. Although AWE numerical models are helpful in informing control

and plant design decisions, different phenomena such as wake effects, vortex shading, added mass effects, and tether/fluid interaction exist in reality, yet are not captured by relatively low-order numerical models. Consequently, it is very desirable for AWE systems to *fuse* numerical modeling tools with inexpensive experimental tools for optimizing system performance. This fusion of plant and controller optimization has been absent in the codesign literature until recently.

In response to the desire to fuse experiments with numerical models in optimizing AWE system designs, the authors of Refs. [21–24] have developed and exploited a lab-scale setup to characterize different AWE plant and control system designs. The system has evolved from a passive test setup at the University of Michigan to a controlled test setup at UNC Charlotte. In the most recent version of this lab-scale setup, AWE lifting body models are three-dimensional (3D) printed and “flown” in a water channel, cameras are used for real-time motion capture, direct current (DC) motors are used to regulate tether lengths, and control algorithms are implemented on a high-performance target computer. The lab-scale AWE prototyping setup at UNC Charlotte enables the fusion of experiments with simulations in the optimization of AWE system controllers and plant designs. Deodhar et al. [25] present an initial codesign result using the lab-scale setup, taking the Altaeros BAT as a case study. The approach followed in Ref. [25] is local in nature. It uses experiments in the vicinity of the estimated optimum design to determine the gradient of performance with respect to design variables, then moves along the direction of that gradient in subsequent iterations. While the initial results of a two-variable optimization in Ref. [25] were positive, this gradient-based approach can easily result in convergence to local optima in the presence of highly nonlinear systems with complex dynamics, especially when more design variables are considered. To address this, our recent conference paper, Ref. [26], proposed an alternative framework that uses optimal design of experiments (DOE) to choose control and plant design variables that span the design space, then uses a statistical quality of fit characterization to shrink the design space at the next iteration, as well as a correction term to improve the numerical model prediction at the next iteration. By ensuring coverage of the design space at each iteration, the optimization approach is much less likely to get stuck in local optima.

While the contribution of Ref. [26] presented a theoretical framework, it lacked actual experimental validation. In this paper, we build upon our initial work done in Ref. [26] by validating the framework with actual experiments and modifying the framework to handle larger numbers of optimization variables. The results presented herein focus on the Altaeros BAT [12] as a case study.

The contributions of this paper are threefold:

- (1) To validate the framework presented in Ref. [26] with physical experiments performed in the UNC-Charlotte lab-scale AWE test setup.
- (2) To extend the numerical model and optimization framework to include additional design variables (specifically, stabilizer areas, which now exist as explicit design variables in the numerical model).
- (3) To validate the effectiveness of the design optimization approach for larger numbers of design variables, using an increased-fidelity model as a surrogate for experiments.

2 Framework for Experimentally Infused Optimization With Iterative Design of Experiment

The work presented herein represents a departure from the vast majority of codesign literature, which assumes that an accurate closed-form representation for the system’s dynamic equations is available. The availability of a closed-form model allows for a variety of numerical tools to be used to optimize the plant and control system design variables, denoted by $\mathbf{p}_p$ and $\mathbf{p}_c$, respectively. Our work, on the other hand, assumes that while a model of the system dynamics exists, it is imperfect. Thus, converging

upon an optimal set of design variables requires fusion of experiments with numerical modeling. The design optimization problem that we are pursuing in this work can be formally summarized as follows:

$$\text{Minimize } J(\mathbf{p}_p, \mathbf{p}_c) = \int_0^{T_f} l(\mathbf{x}(t), \mathbf{u}(t); \mathbf{p}_p, \mathbf{p}_c) dt \quad (1)$$

subject to:

$$\dot{\mathbf{x}} = f_{\text{true}}(\mathbf{x}, \mathbf{u}) \quad (2)$$

$$\mathbf{p}_p \in P \quad (3)$$

$$\mathbf{p}_c \in C \quad (4)$$

Here, $\mathbf{x}$ and $\mathbf{u}$ represent the state and control input vectors, respectively, and Eq. (2) represents the *true* system dynamics, as represented by experimental results. It is assumed that the mathematical structure of $f_{\text{true}}(\mathbf{x}, \mathbf{u})$ is not known, but an approximate numerical model for the dynamics exists. The constraint sets P and C represent hard constraints on the plant and controller design variables, respectively.

Because the true system dynamics are not fully known but instead can only be revealed through experiments, the infusion of numerical modeling tools with experimental results is necessary to address the optimization problem of Eq. (1). In particular, two broad questions may be asked regarding any approach that aims to tackle the optimization problem of Eq. (1) through the fusion of experiments:

- (1) Which experiments should be run in order to reveal the most valuable information for minimizing $J(\mathbf{p}_p, \mathbf{p}_c)$?
- (2) How can the numerical model of the system be improved after each run of experiments in order to facilitate convergence of $\mathbf{p}_p$ and $\mathbf{p}_c$ to their optimal values?

We address the aforementioned questions through an iterative optimization process. This process, termed *experimentally infused optimization*, involves the generation of experiments that span a design space, followed by the reduction of that design space and improvement of the numerical model. The entire process is divided into four main parts as follows:

- (1) Perform an optimal DOE that generates a set of design variables to be characterized through numerical simulations and another set of design variables to be characterized through experiments.
- (2) Characterize a response surface from numerical simulations and experiments.
- (3) Learn from experiments to improve the numerical model for the next iteration.
- (4) Reduce the size of the design space based on the quality of the response surface approximation, then repeat the process.

A flowchart of this process is presented in Fig. 1. Each of the aforementioned processes is explained in detail in this section.

2.1 Optimal Design of Experiment. The overarching objective of DOE is to design experiments over a prescribed design space that maximizes the amount of information gathered by a fixed number of experiments. In this work, G-optimality was used in designing two sets of designs, namely, the numerical simulations and experiments, where N_S and N_E represent the number of simulations and experiments, respectively. In the proposed framework, each of the N_S and N_E simulations and experiments is comprised of a different set of controller *and* plant design variables; hence, the G-optimal design explores the control and plant design spaces *simultaneously*. G-optimal design chooses design points in

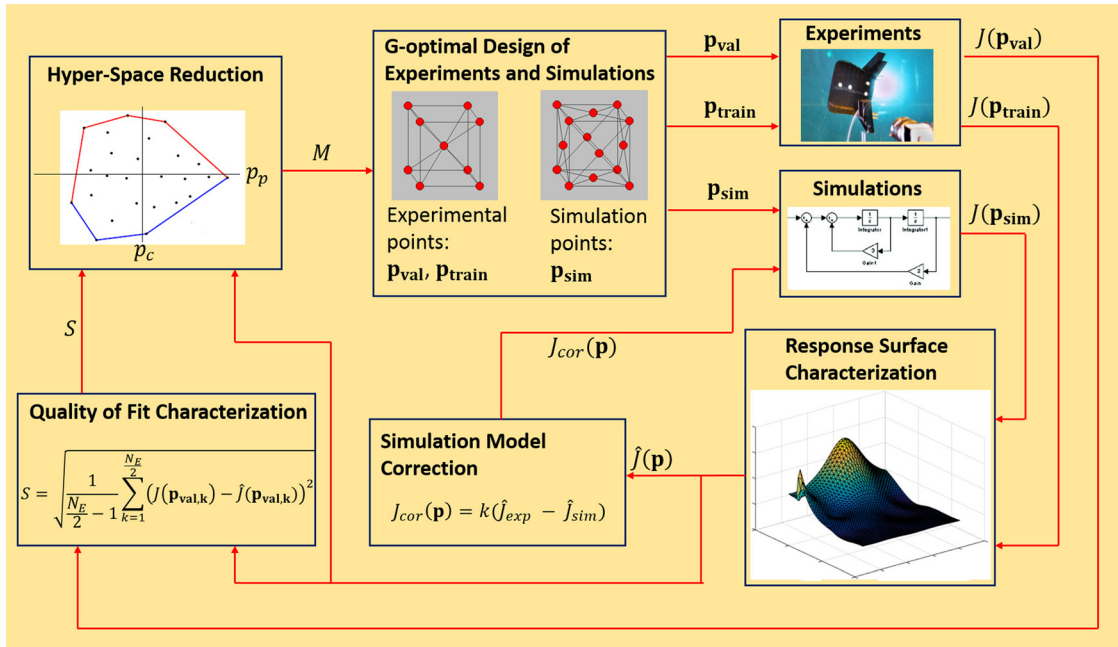


Fig. 1 Flowchart demonstrating the sequence of events involved with an iteration of the experimentally infused optimization process. In short, the process involves the design of experiments, running the simulations/experiments and generating a response surface, learning from the experiments, and reducing the size of the design space for the next iteration.

an attempt to minimize the maximum prediction variance across all experiments. It is known to do an effective job of filling the design space without repeating experiments. G-optimal design relies on the specification of an m element column vector known as the regressor vector, $r(\mathbf{p}_p, \mathbf{p}_c)$, which captures the anticipated dependencies of the system performance on the design variables. For example, for a system with a single scalar plant design variable and control design variable to be optimized, a regressor vector of $r(p_p, p_c) = [p_p \ p_p^2 \ p_c \ p_c^2]^T$ would be chosen if the system designer anticipated that the performance was quadratic with respect to both the plant and controller design variable.

The matrix of n regressor vectors can be expressed as

$$\mathbf{R} = [\mathbf{r}_1 \ \mathbf{r}_2 \ \dots \ \mathbf{r}_n] \quad (5)$$

Here, n corresponds to the number of design points (either numerical simulations or experiments). G-optimality criteria minimize the maximum prediction variance across all the possible design points in the design space, M . The resulting design points are given by

$$[\mathbf{p}_p, \mathbf{p}_c] = \arg \min_{(\mathbf{p}_p, \mathbf{p}_c) \in M} \mathbf{r}_i^T (\mathbf{R}\mathbf{R}^T)^{-1} \mathbf{r}_i \quad (6)$$

Here, $\mathbf{r}_i$ denotes the regressor vector corresponding to the i th design point.

At every iteration, the N_E experimental design points are split into an equal number of training and validation points. In this work, the experimental points were sorted based on their distance from each other, and alternate points were used for training and validation to get better coverage of the design space. The set of training, validation, and simulation points resulting from the G-optimal DOE are denoted by $\mathbf{p}_{\text{train}}$, $\mathbf{p}_{\text{val}}$, and $\mathbf{p}_{\text{sim}}$, respectively.

2.2 Numerical Simulations and Experiments. Experiments and numerical simulations are conducted for every $\mathbf{p}_{\text{train}}$, $\mathbf{p}_{\text{val}}$, and $\mathbf{p}_{\text{sim}}$ using identical operating conditions (i.e., if a controlled perturbation is applied in numerical simulations, then that same controlled perturbation is applied in experiments). An identical

performance index, denoted by $J(\mathbf{p})$, is used to characterize the resulting performance under all experiments and simulations.

2.3 Response Surface Characterization and Numerical Model Correction. Three different response surfaces are characterized at every iteration of the optimization process, specifically:

- (1) Pure experimental response using both training and validation experiments, $\hat{J}_{\text{exp}}$;
- (2) Pure numerical simulations response, $\hat{J}_{\text{sim}}$; and
- (3) Combined experimental training and numerical simulations response, $\hat{J}$.

Each response surface characterization takes the data from experiments or simulations and then uses this data to fit a function over the entire design space. Two techniques are proposed in this work for that response surface characterization.

2.3.1 Parametric Response Surface Characterization Based on a Regressor Vector. In cases where the general structure of the dependencies of performance on the design variables is known, and in cases where the number of design variables is small, it is often possible to relate the performance to the design variables through a regressor vector. Coefficients of the regressor vector can then be determined through weighted least squares regression.

Using this approach, the response surface is estimated using the expression

$$\hat{J}(\mathbf{p}) = \mathbf{r}(\mathbf{p})^T \mathbf{b} \quad (7)$$

where $\hat{J}(\mathbf{p})$ is the estimated performance at any point $\mathbf{p}$, and $\mathbf{b}$ is the vector of regression coefficients obtained by minimizing the weighted sum of squared errors given by

$$\mathbf{b} = \arg \min_{\mathbf{b} \in \mathbb{R}^n} \sum_{i=1}^n w(\mathbf{p}_i) (J(\mathbf{p}_i) - \mathbf{r}(\mathbf{p}_i)^T \mathbf{b})^2 \quad (8)$$

Here, $w(\mathbf{p}_i)$ is the weight on i th point. These weights can be used to value simulations and experiments differently in arriving at the combined response surface, $\hat{J}$.

2.3.2 Distance-Weighted Response Characterization. With a large number of design variables, and in instances when it is difficult to speculate a regressor vector based on known properties of the system, the previous approach may result in poor characterization of the response surface. A more general distance-weighted response surface characterization can yield a more accurate representation of performance over the design space in these cases. In this approach, the performance associated with any point $\mathbf{p}$ is modeled as the weighted sum of all measured points, placing more weight on the points that are closest to $\mathbf{p}$. The response surface is approximated as follows:

$$\hat{J}(\mathbf{p}) = \frac{\sum_{i=1}^n W(\mathbf{p}_i) J(\mathbf{p}_i)}{\sum_{i=1}^n w_i} \quad (9)$$

$$W(\mathbf{p}_i) = \frac{1}{d(\mathbf{p}, \mathbf{p}_i)^2} \quad (10)$$

where $d(\mathbf{p}, \mathbf{p}_i)$ is the Euclidean distance between $\mathbf{p}$ and $\mathbf{p}_i$.

2.4 Numerical Model Improvement. In order to continue to reduce the size of the design space at each iteration, the numerical model must be improved at each iteration. This improvement is accomplished by comparing the response surface generated with experiments to the one generated with the numerical model, then adjusting the numerical model so that it conforms more closely with experimental results at the next iteration. Once the experimental ($\hat{J}_{\text{exp}}$) and numerical ($\hat{J}_{\text{sim}}$) response surface characterizations are obtained, a correction term ($\hat{J}_{\text{cor}}$) is applied to numerical simulations. The form for this correction, at iteration i , is given as follows:

$$\hat{J}_{\text{num}}(\mathbf{p}) = \hat{J}_{\text{sim}}(\mathbf{p}) + \hat{J}_{\text{cor},i}(\mathbf{p}) \quad (11)$$

$$\hat{J}_{\text{cor},i}(\mathbf{p}) = k(\hat{J}_{\text{exp}}(\mathbf{p}) - \hat{J}_{\text{sim}}(\mathbf{p})) + \sum_{j=1}^{i-1} \hat{J}_{\text{cor},j}(\mathbf{p}) \quad (12)$$

Here, $\hat{J}_{\text{cor},i}(\mathbf{p})$ represents the numerical correction at step i , which is obtained by adding a correction from the current iteration to the cumulative correction from all previous iterations (which is reflected in $\hat{J}_{\text{sim}}$ for the current iteration). The variable k , which is taken to be between 0 and 1, limits the correction to a certain percentage of the gap between experiments and the numerical model at each iteration.

2.5 Design Space Reduction. The next goal is to converge close to the optimum design point $\mathbf{p}^*$ by rejecting all candidate configurations that are *decidedly suboptimal*. Determination of what is “decidedly suboptimal” is achieved using hypothesis testing.

The job of the design space reduction is to determine whether one can conclude, for each candidate design within the design space, that the candidate design cannot possibly (within some confidence interval) be the true optimum. If the candidate design cannot possibly be optimal, it is eliminated from the design space; otherwise, it is retained. For these purposes, the null (H_0) and alternate (H_1) hypotheses are defined as

$$H_0 : J(\mathbf{p}) = \hat{J}(\mathbf{p}^*) \quad (13)$$

$$H_1 : J(\mathbf{p}) > \hat{J}(\mathbf{p}^*) \quad (14)$$

The z test is used to reject a candidate configuration by comparing the corresponding z -score to a threshold z_0 for a given confidence interval

$$z(\mathbf{p}) = \frac{\hat{J}(\mathbf{p}) - \hat{J}(\mathbf{p}^*)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (15)$$

$$S = \sqrt{\frac{\sum_{i=1}^{N_E/2} (\hat{J}(\mathbf{p}_{\text{val}}) - J(\mathbf{p}_{\text{val}}))^2}{\frac{N_E}{2} - 1}} \quad (16)$$

The term S is referred to as the *quality of fit* between combined response at validation points and the true experimental response at validation points. n_1 and n_2 denote the number of times each experiment/simulation is repeated.

3 Case Study on an Airborne Wind Energy System

For the case study in this paper, the Altaeros BAT is considered, for which the system model and controller are detailed in this section. This system elevates a horizontal axis turbine to high altitudes through the use of a lighter-than-air annular aerostat. The system is intended to operate in a stationary fashion, and it is essential that the system maintains its altitude and attitude (orientation), while keeping control inputs (tether motions) to a minimum, in order to maximize its energy production.

3.1 Dynamic Model of Airborne Wind Energy System.

The dynamic model of Altaeros system is described in detail in Ref. [22] but is also summarized here for purposes of self-containment. Figure 2 shows the relevant variables used in the dynamic model.

The three tethers are approximated as a single tether with length L_t plus a bridge joint at the top that splits in three attachment points on the BAT. The single tether model circumvents the need to solve differential algebraic equations and instead, results in a set of ordinary differential equations that describe the equations of motion. This effect multiple tethers is approximated by a two degrees-of-freedom bridge joint, whose angular degrees-of-freedom are given by ϕ' and θ' . We refer to ϕ' and θ' as *tether-induced angles*. These angles are approximated by following expressions, which relate the actual individual tether lengths to ϕ' and θ' :

$$\phi' = \tan^{-1} \left(\frac{l_3 - l_2}{l_{\text{sep}}^{\text{lat}}} \right) \quad (17)$$

$$\theta' = \tan^{-1} \left(\frac{l_1 - 0.5(l_2 + l_3)}{l_{\text{sep}}^{\text{long}}} \right) \quad (18)$$

where $l_{\text{sep}}^{\text{long}}$ and $l_{\text{sep}}^{\text{lat}}$ are longitudinal and lateral tether attachment separation distances, respectively. l_1 , l_2 , and l_3 are the lengths of the individual tethers. The center of mass location is a function of Φ (zenith angle), Θ (azimuth angle), and L_t (average tether length). The control inputs are the tether release speeds, $\bar{u}_i$, which are related to the released tether lengths by

$$\frac{d}{dt} l_i = \bar{u}_i \quad (19)$$

The governing equations of motion below are derived using the Euler–Lagrange formulation

$$\mathbf{D}(\mathbf{Q})\ddot{\mathbf{Q}} + \mathbf{C}(\mathbf{Q}, \dot{\mathbf{Q}})\dot{\mathbf{Q}} + \mathbf{g}(\mathbf{Q}) = \boldsymbol{\tau}(\mathbf{Q}, \dot{\mathbf{Q}}, V_{\text{flow}}, \psi_{\text{flow}}) \quad (20)$$

$$\mathbf{X} = \mathbf{f}(\mathbf{Q}, \dot{\mathbf{Q}}) \quad (21)$$

$$\boldsymbol{\Omega} = \mathbf{g}(\mathbf{Q}, \dot{\mathbf{Q}}) \quad (22)$$

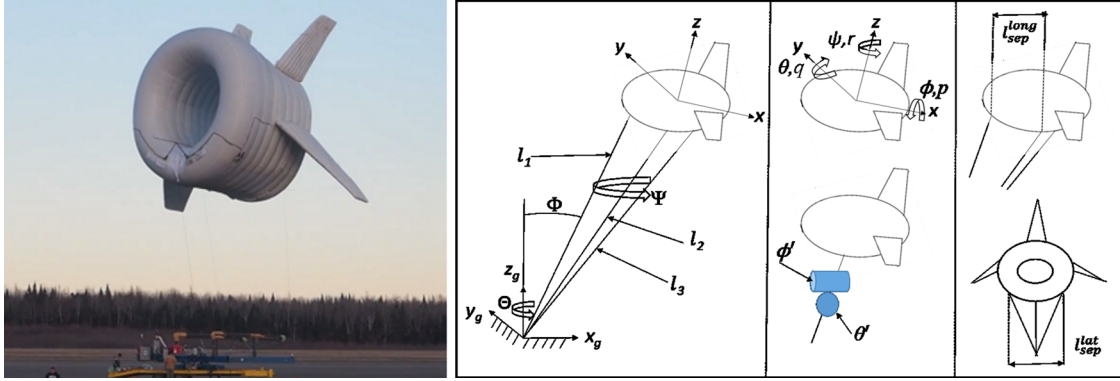


Fig. 2 Left full-scale prototype of an Altaeros BAT used during 2013 flight testing [12]. Right: ground-frame and body-frame coordinates, along with key variables used in deriving equations of motion for the BAT.

where

$$\mathbf{Q} = [\Phi \ \Theta \ \Psi \ L_1 \ \theta' \ \phi'] \quad (23)$$

$$\mathbf{X} = [x_g \ y_g \ z_g \ u \ v \ w] \quad (24)$$

$$\mathbf{\Omega} = [\phi \ \theta \ \psi \ p \ q \ r] \quad (25)$$

Here, V_{flow} is the flow (wind in the case, of the full-scale system, water in the case of the lab-scale water channel-based system) speed, and ψ_{flow} and τ represent flow direction and vector of generalized external forces. Aerodynamic forces and moments are functions of α (angle of attack) and β (side slip angle). Each of the variables in Eq. (23) is summarized in Nomenclature section.

3.2 Closed-Loop Controller Structure. The flight controller for the BAT, first described in Ref. [18], uses the multiple tethers to track three different set points, namely, altitude (z_{sp}), pitch (θ_{sp}), and roll (ϕ_{sp}). Figure 3 presents a block diagram of this controller structure. The controller first computes “virtual” control inputs, denoted by $\bar{v}_z$, $\bar{v}_\theta$, and $\bar{v}_\phi$, which reflect the need to adjust all three tethers synchronously to change altitude, adjust the forward and aft tether asynchronously to change the pitch angle, and adjust the port and starboard tethers asynchronously to adjust the roll angle. These virtual control inputs are the output of filtered proportional plus derivative controllers (lead filters). Because the plant itself contains pure integrators (i.e., a “type 1” system) in the dynamics between the control inputs ($\bar{u}_{\text{center}}$, $\bar{u}_{\text{stbd}}$, and $\bar{u}_{\text{port}}$) and the variables to be controlled (z , θ , and ϕ), integral control is not necessary for tracking in this case, since the plant includes three pure

integrators between the motor speeds and tether lengths. The three control inputs represent tether release speeds, which are controlled in the full-scale system through alternating current induction models and in the lab-scale, water channel-based system by micro DC motors.

The average tether release speed ($\bar{v}_z$), difference between central/peripheral tether release speeds ($\bar{v}_\theta$), and port/starboard tether release speeds ($\bar{v}_\phi$) are related to the tracking errors in altitude (z_e), pitch (θ_e), and roll (ϕ_e) by the transfer functions $G_z(s)$, $G_\theta(s)$, and $G_\phi(s)$, respectively. These transfer functions are given by the following expressions:

$$G_z(s) = \frac{k_d^z s + k_p^z}{\tau_z s + 1} \quad (26)$$

$$G_\theta(s) = \frac{k_d^\theta s + k_p^\theta}{\tau_\theta s + 1} \quad (27)$$

$$G_\phi(s) = \frac{k_d^\phi s + k_p^\phi}{\tau_\phi s + 1} \quad (28)$$

Legacy work by Vermillion et al. (see Ref. [18]) discusses this control structure in more detail, along with mechanisms for calibrating the feedback control parameters. Scaling to the lab-scale environment is discussed in Ref. [22]. Given the symmetrical design of the BAT and the desire for stationary operation, ϕ_{sp} is always equal to zero. The tether release speeds ($\bar{u}_{\text{center}}$, $\bar{u}_{\text{stbd}}$, and $\bar{u}_{\text{port}}$) are related to the virtual control inputs ($\bar{v}_z$, $\bar{v}_\theta$, and $\bar{v}_\phi$) as follows:

$$\mathbf{u} = \mathbf{A} \mathbf{v} \quad (29)$$

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \quad (30)$$

Here, $\mathbf{u}$ represents the vector of control inputs ($\bar{u}_{\text{center}}$, $\bar{u}_{\text{stbd}}$, and $\bar{u}_{\text{port}}$), whereas $\mathbf{v}$ denotes the vector of virtual control inputs ($\bar{v}_z$, $\bar{v}_\theta$, and $\bar{v}_\phi$).

4 Airborne Wind Energy Case Study—Two Design Variables

In this section, we show how the numerical model of Sec. 3 has been fused with lab-scale experiments in order to optimize design variables for the BAT. In this section, we show the results of the experimentally infused optimization framework for two design variables, namely, the longitudinal location of *center of mass* (x_{cm}) and the *trim pitch angle* (θ_{sp}). In Sec. 5, we extend the framework to four design variables.

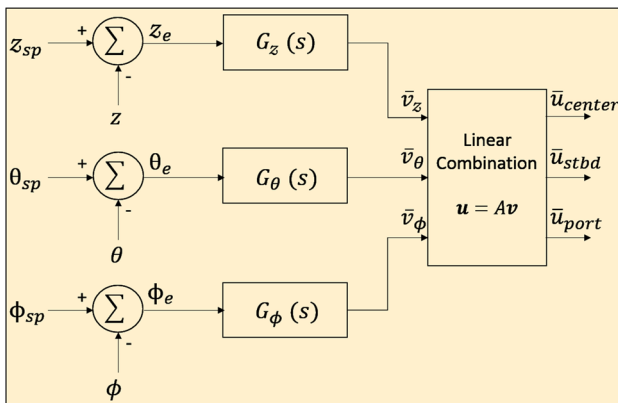


Fig. 3 Block diagram of the flight controller used to track altitude and attitude set points [24]. The present research uses fixed roll and altitude setpoint values.

4.1 Experimental Setup of the Lab-Scale System. Although it is ultimately necessary to validate the plant and control design on a full-scale AWE system, extensive full-scale experiments are expensive and time-consuming. Furthermore, the consequences of a failure are significant, making it inadvisable to explore design configurations that are on the verge of instability for the full-scale system. For these reasons, a lab-scale setup consisting of water channel of $1\text{ m} \times 1\text{ m}$ cross section and 1/100th model of the BAT was developed. This setup evolved from an initial passive system at the University of Michigan (described in Ref. [21]) to the present actively controlled system at UNC Charlotte (initially described in Ref. [23]). The present setup is shown in Fig. 4 and has four major components:

- (1) Three high-speed motion capture cameras for tracking the position and orientation of the BAT,
- (2) DC motors to independently control the lengths of three tethers,
- (3) A high-performance target computer for intense real-time computations and feedback control, and
- (4) A host computer to communicate data between the user and the target computer.

The BAT model is 3D printed out of acrylonitrile butadiene styrene plastic and shown in Fig. 5. A drag screen slit accommodates a mesh screen that replicates the drag properties of a spinning rotor. The center of mass location and net buoyancy can be adjusted by placing lead wires in selected ballast holes. Full position and orientation information is resolved from the location of three sets of white dots on the model, projected onto the field of view of three cameras. Specifically, the cameras capture instantaneous pixel locations of the center of every pair of dots, and, based on the body-frame locations of the dots, the Euler angles and center of mass locations are computed. The technical details of this process are highlighted in Ref. [24].

Finally, a separate dimensional analysis was conducted by the authors in Refs. [21] and [22], demonstrating the dynamic equivalence between the lab-scale and full-scale system, with the exception of uniformly accelerated time constants in the water channel. Specifically, this analysis reveals that, under the assumption of consistent aerodynamic (hydrodynamic) coefficients for the lab-scale and full-scale model (these are not guaranteed due to a Reynolds number mismatch, but have been shown through wind tunnel testing at different Reynolds numbers to be similar), dynamic similarity can be achieved under the following conditions:

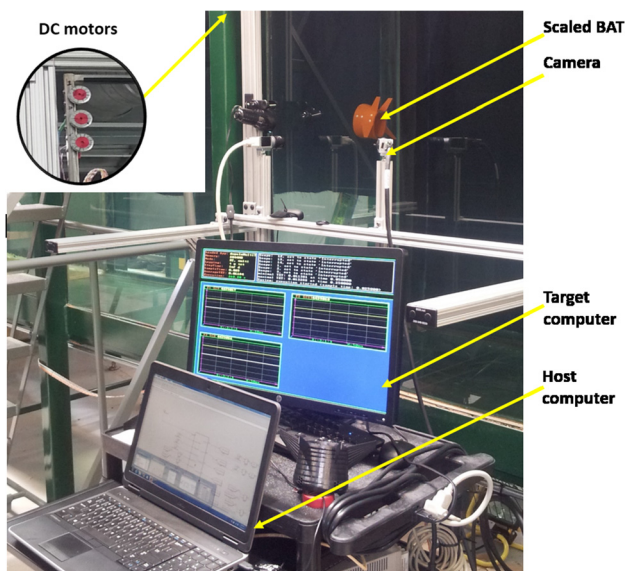


Fig. 4 Water channel-based test platform at UNC Charlotte

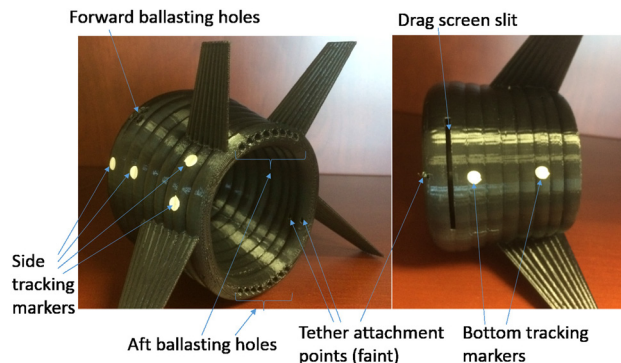


Fig. 5 Isometric and bottom views of the 1/100-scale BAT model used for testing. The ballast holes provide the ability to modify the center of mass location between iterations.

- (1) Maintaining *geometric similarity* between the lab-scale and full-scale system (requires all lengths to be scaled according to L_f),
- (2) Maintaining a consistent *net buoyancy* between the lab-scale and full-scale system (requires the quantity $m/\rho L_f^3$ to be maintained between the lab-scale and full-scale system), and
- (3) Maintaining a consistent *Froude number* between the lab-scale and full-scale system (requires the flow velocity, V_{flow} to be proportional to $\sqrt{L_f}$).

Following these scaling rules, with consistent aerodynamic (hydrodynamic) coefficients between scales, the dynamics of the lab-scale system will be dynamically similar to those of the full-scale systems, with time constants that are accelerated according to the square root of the length scale ($\tau \propto \sqrt{L_f}$).

4.2 Performance Index. Optimizing the performance of the BAT via the techniques of Sec. 2 requires us to *quantify* performance. For the BAT, this was done through a *cost function*, J , which we desire to minimize. The cost function used for this case included six terms that penalize tracking errors (on roll, pitch, and altitude), ground footprint (using zenith angle) and energy consumption (through the control inputs). The cost function structure is given as follows:

$$J = \int_{T_i}^{T_f} (k_1(\psi - \psi_{\text{flow}})^2 + k_2(\theta_e)^2 + k_3(\phi_e)^2 + k_4(z_e)^2 + k_5(\mathbf{u}^T \mathbf{u}) + k_6(\Phi)^2) dt \quad (31)$$

Here, $k_1, \dots, k_6$ are the weights on each term in the cost function. These weights were chosen to emphasize some components of the cost over other and account for the different engineering units in each of the terms.

4.3 Experimental Profile and Perturbations. Every training and validation experiment was conducted using identical flow speeds and lateral perturbations to the ones used in numerical simulations. Specifically, every experiment/simulation was conducted for a period of 300 s, wherein three different flow speeds of 0.205, 0.245, and 0.285 m/s were used over each 100 s interval. Twenty seconds after each speed change, the BAT was subjected to a sequence of lateral perturbations, which were induced by pulling down on the port tether at full speed for 1 s, followed by a 1 s delay before resuming closed-loop control. This perturbation profile can be represented as follows:

$$\bar{u}_{\text{port}} = \begin{cases} -u_{\text{max}}, & \text{if } t \in T \\ \bar{u}_z - \bar{u}_\theta - \bar{u}_\phi, & \text{otherwise} \end{cases} \quad (32)$$

$$T = [20\text{ s } 22\text{ s}] \cup [120\text{ s } 122\text{ s}] \cup [220\text{ s } 222\text{ s}] \quad (33)$$

Table 1 Table showing estimated optimal design variables (values of p^*) at each iteration, with the zeroth iteration corresponding to the purely numerically optimized design

Iteration	0	1	2	3
x_{cm} (percentage of chord length)	45.7	49.40	49.53	49.53
θ_{sp} (deg)	8.29	7.21	5.45	7.51
Area of feasible design space (percentage of original design space)	100	61.53	16.52	15.33

4.4 Case Study Results. The framework in Sec. 2 was used to optimize center of mass location and the trim pitch angle. The weights $w(\mathbf{p})$ given by Eq. (8) were chosen as follows: training experiments $w(\mathbf{p}_{train})$ had a weight of 80% as compared to only 20% for numerical simulations $w(\mathbf{p}_{sim})$ in the response surface characterization. For response surface characterization, we used a quadratic response surface for which the regressor vector is given by

$$\mathbf{r} = [1 \quad p_p^2 \quad p_p \quad p_c^2 \quad p_c]^T \quad (34)$$

A 90% confidence interval was used for determining whether or not to reject points from the design space based on the z-score of Eq. (15).

The results of three iterations of the experimentally infused optimization process, along with the optimal configuration given purely by the numerical model, are presented in Table 1. The left half of Fig. 6 shows the G-optimal experiments and numerical simulations generated at every iteration, and the right half shows the corresponding $\hat{J}$ at for every combination of design variables in the new design space. It is evident from the table and figures that the design space is shrinking between every iteration. Thereafter, the variation in experiments is dictated by the noise in the experimental data, which limits further shrinking of the design space. The shrinking at very first iteration is not as significant as that of the second iteration since the numerical model remains uncorrected during the first iteration. At the onset of second iteration, the numerical model is corrected using experiments, which leads to a better quality of fit between combined numerical plus training data and the validation data. Figure 7, which compares the quality of flight under purely numerical optimization to the quality of flight after three iterations of the experimentally infused process, shows a significant improvement as a result of the incorporation of experiments.

5 Extended Optimization Framework for Four Design Variables

Figure 7 shows that a notable improvement in the AWE system performance results from the optimization of just two design variables. However, to further improve the system response, this section shows how two design variables can be added to the previous ones to improve the performance even more. In particular, this section focuses on the optimization of four design parameters, namely, the longitudinal center of mass location, trim pitch angle, horizontal stabilizer area, and vertical stabilizer area. Thus, the additional design variables chosen were:

- (1) Horizontal stabilizer surface areas (A^H) and
- (2) Vertical stabilizer surface areas (A^V)

The pairs of horizontal and vertical stabilizers are critical in reducing heading oscillations. In order to characterize the impact of varying stabilizer surface areas, the model presented in Sec. 3 was modified such that the aerodynamic force and moment coefficients for the overall body were partitioned between the main body (the “fuselage”) and the stabilizers. These equations in the modified numerical model are given by

$$C_{D,L,S}(\alpha, \beta) = C_{D,L,S}^F(\alpha, \beta) + C_{D,L,S}^H(\alpha, \beta) \frac{A^H}{A_{ref}} + C_{D,L,S}^V(\alpha, \beta) \frac{A^V}{A_{ref}} \quad (35)$$

$$C_{M_x, M_y, M_z}(\alpha, \beta) = C_{M_x, M_y, M_z}^F(\alpha, \beta) + C_{M_x, M_y, M_z}^H(\alpha, \beta) \frac{A^H l^H}{A_{ref} l_{ref}} + C_{M_x, M_y, M_z}^V(\alpha, \beta) \frac{A^V l^V}{A_{ref} l_{ref}} \quad (36)$$

where C_D , C_L , and C_S are the drag, lift, and side force coefficients, whereas C_{M_x} , C_{M_y} , and C_{M_z} are the roll, pitch, and yaw moment coefficients. α and β are angles of attack and the side slip angle, and A_{ref} and l_{ref} are the reference area and reference length for the full system. The variables A^H and A^V represent the horizontal and vertical stabilizer surface areas, respectively, which serve as design variables to be optimized. The parameters l^H and l^V represent reference lengths for the horizontal and vertical stabilizers, respectively. These parameters were adjusted such that a constant aspect ratio was maintained for any combination of stabilizer surface areas (A^H and A^V , respectively).

5.1 Numerical Alternative to Experiments. For our initial optimization of four design variables, we used an increased-fidelity model as a *surrogate* for expensive experiments. The alternate model incorporated stall effects for the fuselage and the horizontal stabilizer that were omitted from the numerical model. Previous wind tunnel testing on a 1/20th scaled BAT model in Ref. [21] showed that the body and horizontal stabilizers would stall beyond a particular angle of attack. To replicate these stall effects, the lift coefficients on body and stabilizer were modeled as follows:

$$C_L^F = \begin{cases} C_{L_0}^F + C_{L_x}^F \alpha^F, & \text{if } |\alpha^F| \leq \alpha_{stall}^F \\ C_{L_0}^F + 2C_{L_x}^F \alpha_{stall}^F - C_{L_x}^F \alpha^F, & \text{otherwise} \end{cases} \quad (37)$$

$$C_L^H = \begin{cases} C_{L_0}^H + C_{L_x}^H \alpha^H, & \text{if } |\alpha^H| \leq \alpha_{stall}^H \\ C_{L_0}^H + 2C_{L_x}^H \alpha_{stall}^H - C_{L_x}^H \alpha^H, & \text{otherwise} \end{cases} \quad (38)$$

Here, $C_L^F(\alpha^F)$ and $C_L^H(\alpha^H)$ denote the lift coefficients for main body and the horizontal stabilizer, respectively; $C_{L_0}^F$ and $C_{L_0}^H$ are the main body and horizontal stabilizer lift coefficients at zero angle of attack; $C_{L_x}^F$ and $C_{L_x}^H$ represent the linear lift coefficients; α^F and α^H are the angles of attack; and α_{stall}^F and α_{stall}^H are the angles of attack at which the main body and horizontal stabilizers stall.

5.2 Results Using Alternative Numerical Model as Experiments. The experimentally infused optimization framework was applied to modified numerical model and surrogate experiments using identical test profile as in the case of a two-variable optimization. Unlike previous weights of 80/20 on experiments versus numerical simulations, the weight on the experiment was reduced to 60% since the actual experiments were replaced by a surrogate model. The results are shown in Table 2,

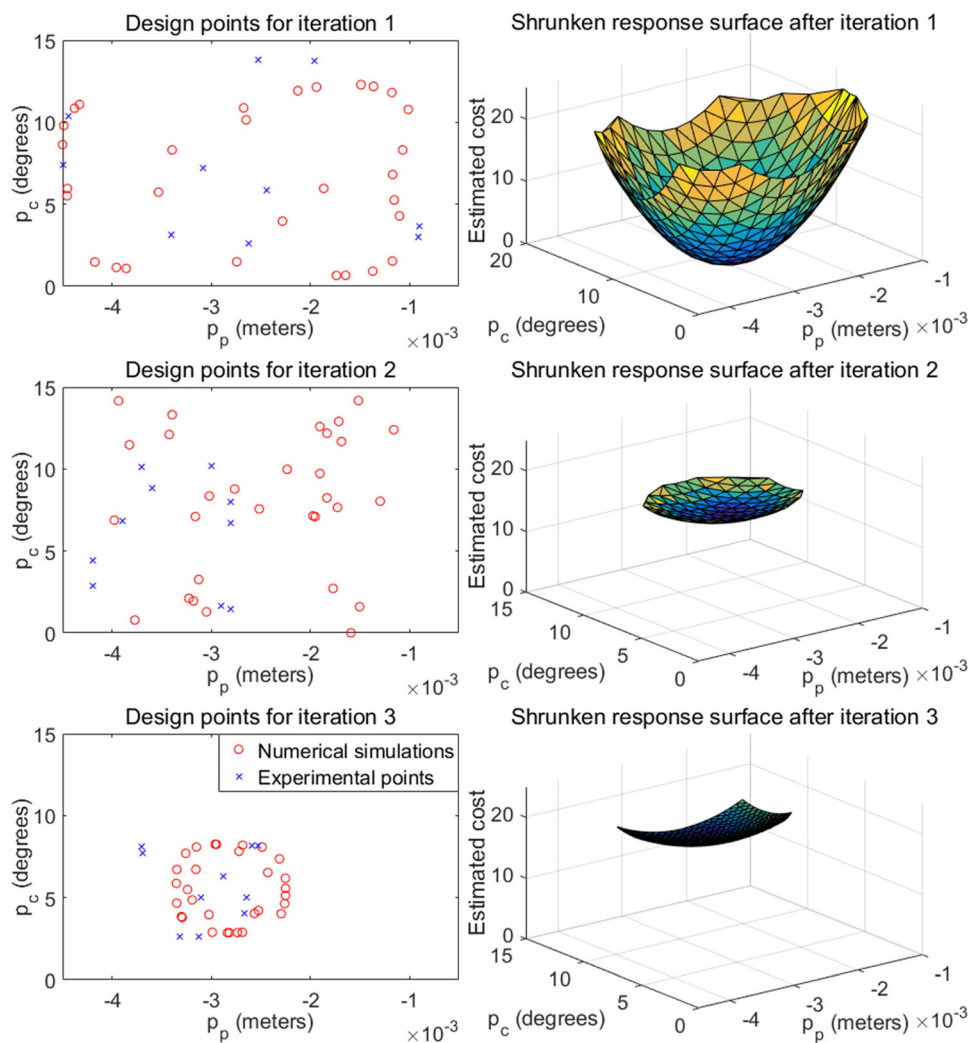


Fig. 6 The experiments and numerical simulation points generated by G-optimal design at every iteration are shown in the left half. The right half shows surface plots of combined numerical and experimental response surface (J) for every (p_p, p_c) combination in the shrunken design space.

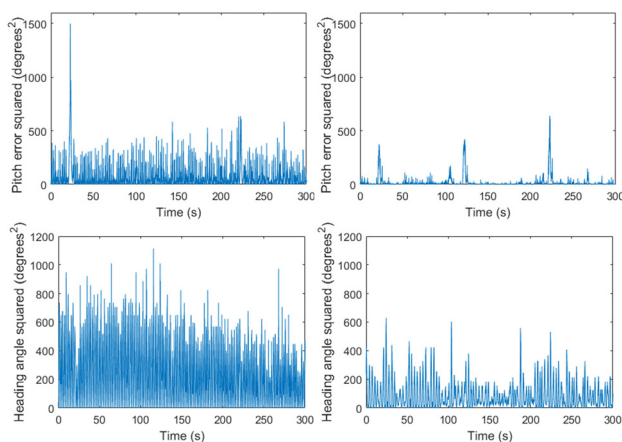


Fig. 7 The left and right halves of this figure show a comparison between the numerically optimal configuration and experimentally infused optimum, respectively. A pronounced improvement in pitch angle and heading angle error with experimental infusion is evident from the plots.

where performance is improved at each iteration and the size of the design space is reduced. The zeroth iteration represents the results of numerical optimization alone. The comparison of system responses for optimal design given at zeroth iteration (i.e., the purely numerically optimized system) and third iteration is shown in Fig. 8. The figure shows that the purely numerically optimized design yields a limit-cycle behavior after the first perturbation. It can be seen that the experimentally infused optimization process produces a design that performs significantly better than the design that is optimized based on the simple numerical model alone. A tradeoff between quality of flight given by tracking errors and ground footprint given by the zenith angle is observed in the case of four-variable optimization (also seen in Fig. 9). The quality of flight is improved greatly at the cost of a slight increase in the ground footprint.

6 Conclusions and Future Work

A formal iterative method for combined plant and controller optimization, fusing both simulations and experiments, was proposed in this research. The framework was validated using an AWE application, initially using the center of mass location and pitch angle set point as optimization variables. By applying the experimentally infused optimization process, using a unique lab-scale setup for the experiments themselves, we demonstrated that

Table 2 Table showing estimated optimal design variables (values of p^*) at each iteration

Iteration	0	1	2	3
x_{cm} (percentage of chord length)	46.3	46.82	50.59	49.03
θ_{sp} (deg)	12.73	1.25	5.67	4.85
$A^H \times 10^{-3} \text{ m}^2$	1.8	2.8	2.6	2.0
$A^V \times 10^{-3} \text{ m}^2$	2.5	1.8	2.0	2.5
Hyper volume of feasible design space (percentage of original design space)	100	65.95	54.67	36.92

Note: Iteration zero indicates optimum design variables obtained from pure numerical simulations.

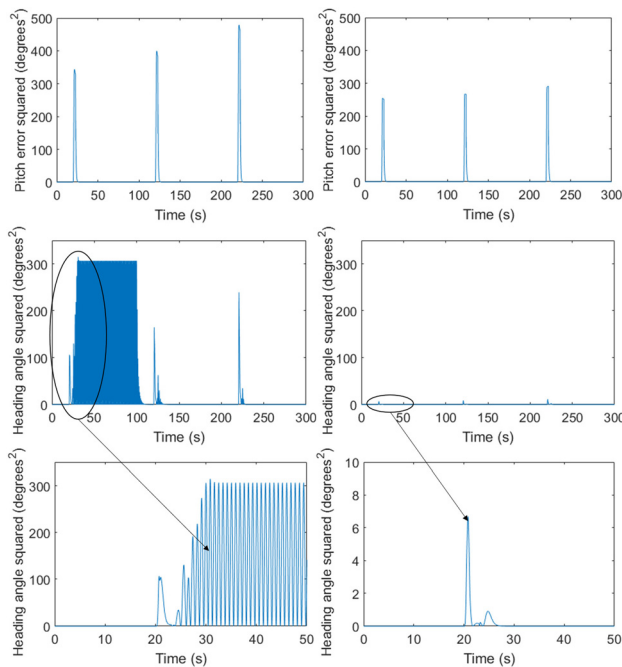


Fig. 8 This figure shows a comparison between pitch angle and heading angle responses for the numerically optimized design (left) versus the design resulting from the experimentally infused optimization approach (right)

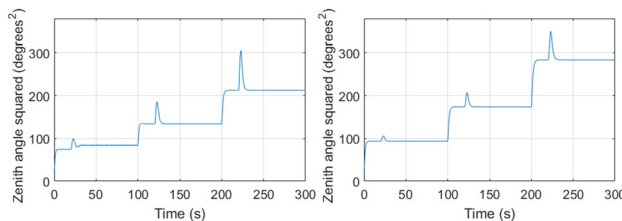


Fig. 9 The experimentally infused optimum results in a slightly higher zenith angle than the numerically optimized design. Thus, the improvement in flight quality seen in Fig. 8 comes at the (relatively minor) cost of an increase in ground footprint.

we can obtain much better system performance through the infusion of experiments than with the numerical model alone. We also extended the results of our framework to include additional design variables, adding horizontal and vertical stabilizer surface areas to the list of design variables to be optimized. We performed an initial demonstration of the effectiveness of our approach for larger numbers of design variables (four, in this case), using a higher-fidelity “surrogate experimental” model in place of experiments. Future work will replace the surrogate experimental model with actual experiments using rapidly reconfigurable 3D printed models that enable us to “plug-and-play” different stabilizer geometries. In addition, the controller set points will be optimized in real time based on flow conditions and power production. The stability

and robustness of the experimentally infused framework will be assessed through different convergence criteria.

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Nomenclature

- L_t = average tether length
- p = angular velocity about body fixed x axis
- q = angular velocity about body fixed y axis
- r = angular velocity about body fixed z axis
- u = body frame x velocity
- v = body frame y velocity
- w = body frame z velocity
- x_g = ground frame x position
- y_g = ground frame y position
- z_g = ground frame z position
- θ = Euler pitch angle
- Θ = azimuth angle
- θ' = tether-induced pitch angle
- ϕ = Euler roll angle
- ϕ' = tether-induced roll angle
- ψ = Euler yaw angle
- Ψ = zenith angle
- Ψ = “twist” angle (around tether)

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